WORKED SOLUTION

STEP II 2023, Question 6

Pure • Fibonacci Matrix Identities

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Once the Fibonacci numbers are packaged as the entries of the powers of a single 2×2 matrix, every identity about them becomes an identity about powers of $\mathbf{Q}$, where ordinary algebra (index laws, determinants, the binomial theorem) does the work. Each part follows the same recipe: find two ways of writing the same matrix, then read off one entry, almost always the top-right one, which is F_n itself.

Solution

The matrix formula

For $n = 1$ the left-hand side is $\begin{pmatrix} F_2 & F_1 \\ F_1 & F_0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \mathbf{Q}$, since $F_2 = F_1 + F_0 = 1$. So the result holds for $n = 1$.

Suppose it holds for some $n \geq 1$. Then

$$\mathbf{Q}^{n+1} = \mathbf{Q}^n \mathbf{Q} = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} F_{n+1} + F_n & F_{n+1} \\ F_n + F_{n-1} & F_n \end{pmatrix} = \begin{pmatrix} F_{n+2} & F_{n+1} \\ F_{n+1} & F_n \end{pmatrix},$$

using the recurrence twice. This is the formula with n replaced by $n+1$, so by induction it holds for all positive integers n.

Part (i)

Determinants are multiplicative, so $\det(\mathbf{Q}^n) = (\det \mathbf{Q})^n = (0 - 1)^n = (-1)^n$. Computing the same determinant from the entries,

$$\det(\mathbf{Q}^n) = F_{n+1}F_{n-1} - F_n^2,$$

and therefore $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$, as required. (This is Cassini's identity.)

Part (ii)

By the index law, $\mathbf{Q}^{m+n} = \mathbf{Q}^m\mathbf{Q}^n$:

$$\begin{pmatrix} F_{m+n+1} & F_{m+n} \\ F_{m+n} & F_{m+n-1} \end{pmatrix} = \begin{pmatrix} F_{m+1} & F_m \\ F_m & F_{m-1} \end{pmatrix}\begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}.$$

The top-right entry of the product is the first row of $\mathbf{Q}^m$ times the second column of $\mathbf{Q}^n$, namely $F_{m+1}F_n + F_mF_{n-1}$. Comparing top-right entries gives

$$F_{m+n} = F_{m+1}F_n + F_mF_{n-1}.$$

Part (iii)

Directly,

$$\mathbf{Q}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \mathbf{I} + \mathbf{Q}.$$

(This is the matrix version of $x^2 = 1 + x$, the equation satisfied by the golden ratio.)

In the parts below, note that the top-right entry of $\mathbf{Q}^k$ is F_k for every $k \geq 1$, and also for $k = 0$, since $\mathbf{Q}^0 = \mathbf{I}$ has top-right entry $0 = F_0$. Taking the top-right entry of a sum of matrices is linear, so the top-right entry of $\sum c_k\mathbf{Q}^k$ is $\sum c_kF_k$.

(a) Using $\mathbf{Q}^2 = \mathbf{I} + \mathbf{Q}$ and the binomial theorem with $\mathbf{A} = \mathbf{Q}$,

$$\mathbf{Q}^{2n} = (\mathbf{Q}^2)^n = (\mathbf{I} + \mathbf{Q})^n = \sum_{k=0}^{n}\binom{n}{k}\mathbf{Q}^k.$$

The top-right entry of the left side is F_{2n}, and that of the right side is $\sum_k \binom{n}{k}F_k$. Hence $F_{2n} = \sum_{k=0}^{n}\binom{n}{k}F_k$.

(b) First find $\mathbf{Q}^3$ in the same style:

$$\mathbf{Q}^3 = \mathbf{Q}\,\mathbf{Q}^2 = \mathbf{Q}(\mathbf{I} + \mathbf{Q}) = \mathbf{Q} + \mathbf{Q}^2 = \mathbf{Q} + \mathbf{I} + \mathbf{Q} = \mathbf{I} + 2\mathbf{Q}.$$

Now apply the binomial theorem with $\mathbf{A} = 2\mathbf{Q}$:

$$\mathbf{Q}^{3n} = (\mathbf{I} + 2\mathbf{Q})^n = \sum_{k=0}^{n}\binom{n}{k}2^k\mathbf{Q}^k,$$

and comparing top-right entries gives $F_{3n} = \sum_{k=0}^{n}\binom{n}{k}2^kF_k$.

For the second identity, write $\mathbf{Q}^{3n} = \mathbf{Q}^n\mathbf{Q}^{2n}$ and use the expansion from (a):

$$\mathbf{Q}^{3n} = \mathbf{Q}^n\sum_{k=0}^{n}\binom{n}{k}\mathbf{Q}^k = \sum_{k=0}^{n}\binom{n}{k}\mathbf{Q}^{n+k}.$$

The top-right entry of $\mathbf{Q}^{n+k}$ is F_{n+k}, so $F_{3n} = \sum_{k=0}^{n}\binom{n}{k}F_{n+k}$.

(c)

Ask yourself Which matrix should we raise to the nth power to produce the sign $(-1)^{n+k}$?

The sign $(-1)^{n+k} = (-1)^{n-k}$ is what appears when we expand $(\mathbf{Q}-\mathbf{I})^n$. To fit the form of the binomial theorem we are allowed to use, write $\mathbf{Q}-\mathbf{I} = -(\mathbf{I}+(-\mathbf{Q}))$, so

$$(\mathbf{Q}-\mathbf{I})^n = (-1)^n(\mathbf{I}+(-\mathbf{Q}))^n = (-1)^n \sum_{k=0}^{n} \binom{n}{k}(-1)^k\mathbf{Q}^k = \sum_{k=0}^{n}(-1)^{n+k}\binom{n}{k}\mathbf{Q}^k.$$

Multiplying by $\mathbf{Q}^n$,

$$\sum_{k=0}^{n}(-1)^{n+k}\binom{n}{k}\mathbf{Q}^{n+k} = \mathbf{Q}^n(\mathbf{Q}-\mathbf{I})^n = (\mathbf{Q}(\mathbf{Q}-\mathbf{I}))^n,$$

where the last step is allowed because $\mathbf{Q}$ and $\mathbf{Q}-\mathbf{I}$ commute (both are polynomials in $\mathbf{Q}$), so the factors can be regrouped. But by (iii), $\mathbf{Q}(\mathbf{Q}-\mathbf{I}) = \mathbf{Q}^2-\mathbf{Q} = \mathbf{I}$, so the whole sum equals $\mathbf{I}^n = \mathbf{I}$.

The top-right entry of $\mathbf{I}$ is 0, while that of the left side is $\sum_k(-1)^{n+k}\binom{n}{k}F_{n+k}$. Therefore

$$\sum_{k=0}^{n}(-1)^{n+k}\binom{n}{k}F_{n+k} = 0.$$

As a bonus, the top-left entries give $\sum_k(-1)^{n+k}\binom{n}{k}F_{n+k+1} = 1$.

Checks

- (i) with $n=5$: $F_6F_4 - F_5^2 = 8\cdot 3 - 25 = -1 = (-1)^5$.
- (ii) with $m=3$, $n=4$: $F_4F_4 + F_3F_3 = 9+4 = 13 = F_7$.
- (a) with $n=4$: $0+4\cdot 1+6\cdot 1+4\cdot 2+1\cdot 3 = 21 = F_8$.
- (b) with $n=3$: $3\cdot 2\cdot 1+3\cdot 4\cdot 1+8\cdot 2 = 34 = F_9$, and also $F_3+3F_4+3F_5+F_6 = 2+9+15+8 = 34$.
- (c) with $n=3$: $-F_3+3F_4-3F_5+F_6 = -2+9-15+8 = 0$.

All five identities were also confirmed by computer for every n up to 59 (and every m, n up to 49 in (ii)).

Common slips

- In (ii), comparing the wrong entry, or multiplying a row by a row, gives a plausible-looking but false formula. The top-right entry of $\mathbf{Q}^m\mathbf{Q}^n$ uses the first *row* of $\mathbf{Q}^m$ and the second *column* of $\mathbf{Q}^n$.
- Using the binomial theorem for $(\mathbf{A}+\mathbf{B})^n$ with matrices that do not commute. Here every matrix is a polynomial in $\mathbf{Q}$, so everything commutes, but it is worth saying so.

- Forgetting the $k = 0$ term: it contributes $\mathbf{I}$, whose top-right entry is $F_0 = 0$.

Beyond STEP

The relation $\mathbf{Q}^2 = \mathbf{I} + \mathbf{Q}$ is the Cayley–Hamilton theorem for $\mathbf{Q}$: every square matrix satisfies its own characteristic equation, here $\lambda^2 - \lambda - 1 = 0$. Diagonalising $\mathbf{Q}$, whose eigenvalues are $\frac{1}{2}(1 \pm \sqrt{5})$, gives Binet's closed formula for F_n, and the matrix powers also give a fast way to compute F_n by repeated squaring. Strang's *Introduction to Linear Algebra* treats Fibonacci numbers in exactly this way when it introduces eigenvalues.

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